



Vietnamese-German University

Vietnamese-German University
Computer Science Study Program

61CSE326 – Project

Collaborative AMR for Warehouse-to-Aisle Stock Replenishment

Weekly Report 01

by

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(September 15, 2026)

Disclaimer

We declare that this report is a product of our own work, unless otherwise referenced. We also declare that all opinions, results, conclusions, and recommendations are our own and may not represent the policies or opinions of the Computer Science study program of the Vietnamese-German University.

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1 Introduction

1.1 Project Context & Identification

In autonomous warehouse, mobile robot faces an existing challenge: navigating around nondeterministic obstacles, including misplaced storage pallets, and dynamic objects such as walking human workers and moving carts.

Under the supervision of Dr.-Ing. Quang Huan Dong, this project addresses **Use Case 6 (UC6)**: *A mobile robot navigates to storage locations, retrieves specified items from shelves, and transports them to designated drop-off points.*

1.2 Problem Statement & Operational Scope Definition

The Commercial Restocking Challenge: In modern retail superstores, restocking goods from backroom inventory to customer aisles takes a massive amount of physical labor. Staff members spend hours walking miles pushing heavy manual carts.

Scope Reduction - Avoiding Front-of-House Shelf Facing: Attempting to have a robot place individual products face-forward directly onto retail shelves requires high-degrees-of-freedom manipulator arms, soft grippers, and complex 3D computer vision. In industrial robotics, this is fragile, costly, and would divert focus away from the core use-case idea of this project. Therefore, this is not included in the scope of our project.

Our Proposed Collaborative Scope (Warehouse-to-Aisle Logistics): We adapt UC6 into a realistic **Human-Robot Collaborative Replenishment System**:

1. The Autonomous Mobile Robot (AMR) navigates to backroom storage racks, docks under bulk totes or mini-pallets, and raises them using a vertical lift mechanism.
2. The AMR moves through corridors and transports cargo to **Designated Staging Locations** (e.g., the ends of retail aisles), as mapped in Figure 1.
3. At these designated locations, human retail staff (or specialized pick-and-place robots) unload the bulk goods and perform the shelf-facing of organize the individual items on the shelves (see Figure 2).

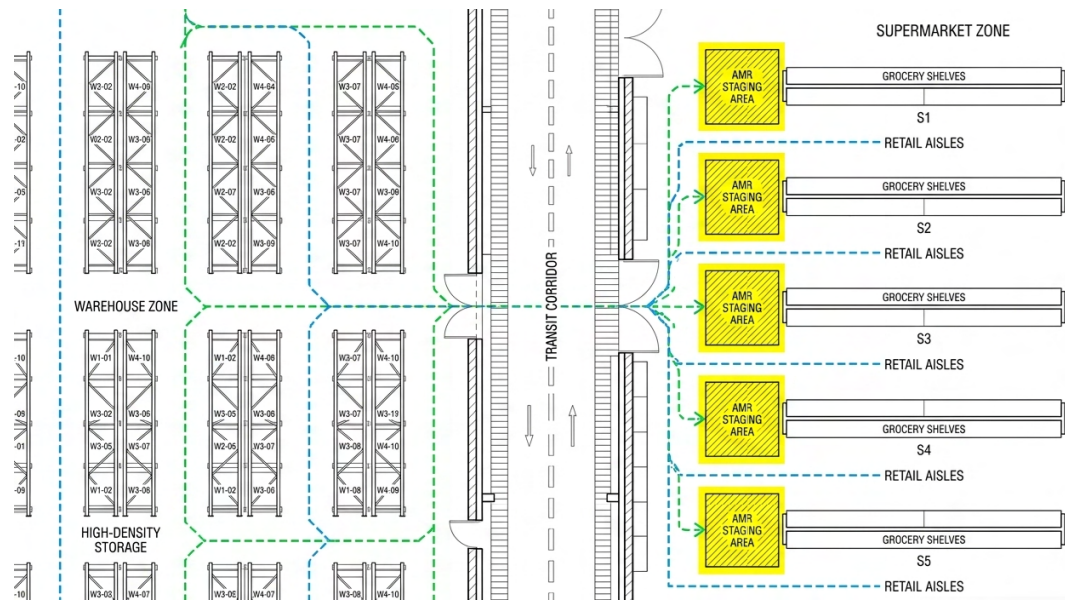


Figure 1: The operational map layout. The left zone highlights warehouse storage aisles, connected to the retail supermarket area on the right with yellow AMR staging locations.

Social Value: Shifting repetitive heavy loads to an autonomous vehicle relieves human workers of intense physical strain and fatigue. As illustrated in Figure 2, this opens up warehouse and inventory jobs to a much wider, more inclusive pool of people, including older workers or persons with physical constraints.

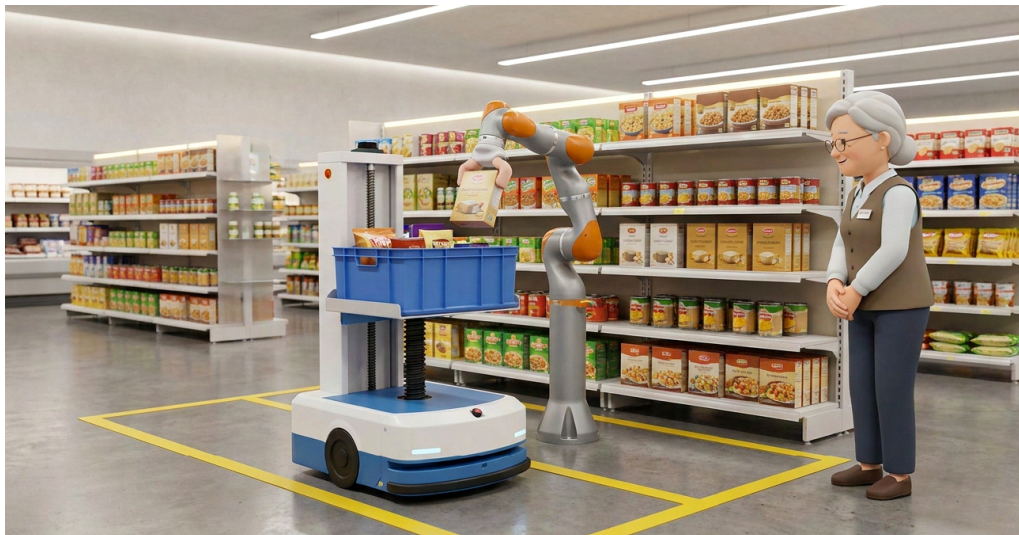


Figure 2: The supermarket aisle staging area. The AMR docks within floor-marked boundaries, where an elderly worker collaborates alongside a secondary pick-and-place arm transferring items onto shelves.

Core Problem Formulation: Operating across shared commercial corridors requires the path-planning software to handle:

1. **Static Obstacle Avoidance:** Safely maneuvering around dropped packaging boxes, misplaced carts, and pillars.
2. **Dynamic Obstacle Avoidance:** Detecting, slowing down, or replanning around moving pedestrians and crossing carts in shared aisles.

2 Methods & Technical Investigation

2.1 System Architecture & Simulation Pipeline

To establish a scalable digital twin environment, we are evaluating a hybrid architecture pairing **Unity (Unity 6)** with **ROS 2 Humble Hawksbill**.

2.2 Robot Structure & Mechanism Concept

Our proposed robot consists of two simple parts:

1. **Two-Wheel Mobile Base:** The robot moves using two motorized wheels (differential drive) and small support caster wheels. This lets the robot turn in place smoothly on flat indoor floors.
2. **Vertical Lift Mechanism:** Instead of a complex robotic arm with many moving joints, the robot uses a simple elevator-style lift (moving up and down by 10 to 30 cm). It raises to pick up a storage box and lowers to drop it off, controlled by a simple ROS 2 command (`/lift_cmd`).



Figure 3: Collaborative Stacker AMR physical layout.

2.3 Virtual-to-Hardware Framework

To ensure future physical hardware deployment remains feasible without rewriting our core logic (subject to timeline and resources):

- Restricting core navigation interactions strictly to standard ROS 2 message definitions.
- **Hardware Abstraction Planning:** Researching an embedded bridge path using **micro-ROS** on microcontrollers (**STM32 Nucleo** / **ESP32**) to replace the virtual Unity TCP socket when hardware prototyping begins.

3 Results (Week 1 Progress)

During the initial kickoff week, the following preliminary setup tasks were completed:

1. **Scope & Use Case Formulation:** Defined the Collaborative Warehouse-to-Aisle Stacker AMR use case, establishing clear boundaries between autonomous transport and human shelf stocking.
2. **Development Environment Setup:**
 - Configured native Ubuntu 22.04 LTS + ROS 2 Humble environments on team laptops.
 - Installed Unity Hub, Unity 6 LTS.
3. **Version Control:** Initialized the team GitHub repository, configured workspace directories, and prepared the weekly reporting structure.

4 Discussion & Upcoming Week Roadmap

4.1 Discussion of Scope

By deliberately focusing on transport with a "1 Degree of Freedom" lift rather than fine-grained arm manipulation, we mitigate the risk of getting slowed down with complex movement while developing. This ensures that the majority of our development effort remains dedicated to the core course requirement: **static and dynamic obstacle avoidance planning**.

4.2 Action Plan for Upcoming Week (Sprint 1)

To prepare for Milestone 1 (Session 5: Demo tutorial adaptation), our goals for Week 2 include:

- **Self Educate:** Research and find tutorials on ROS 2 fundamental concepts and virtualization methods with Unity.
- **ROS-TCP Bridge Verification:** Launch `ros_tcp_endpoint` server in ROS 2 and establish bidirectional loopback with Unity.
- **URDF Staging:** Import a baseline mobile stacker URDF model into Unity to test joint articulation and physics.
- **Teleoperation Testing:** Verify basic chassis motion in Unity by publishing commands from ROS 2.
- **Environment Layout:** Set up preliminary warehouse storage racks and aisle drop-off markers in the 3D scene.